FOUR EQUIVALENT CHARACTERISATIONS OF A NOETHERIAN AFFINE FORMAL ALGEBRAIC SPACE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file formal-spaces.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- Affine formal algebraic spaces as formal spectra, and the implications between their types. Same chapter, Definition definition-affine-formal-algebraic-space used in the statement and Lemma lemma-implications-between-types, invoked at proof lines 2584-2586. Supply the presentations and the two immediate implications.
- A closed subscheme of an affine formal algebraic space over a McQuillan one is affine and closed in the spectrum. Same chapter, Lemma lemma-mcquillan-affine-formal-algebraic-space, invoked at proof lines 2577-2579. Gives the first implication.
- A completion of a Noetherian adic ring, and the criterion for a ring to be Noetherian. Algebra, Lemma algebra-lemma-completion-Noetherian, invoked at proof lines 2595-2597. Gives the step from the third description to Noetherianness.
- A weakly adic ring is admissible with an ideal of definition whose square closure is open, and finite generation of an ideal of definition. Same chapter, Lemma lemma-weakly-adic at proof lines 2601-2604, and Lemma lemma-weakly-adic-finite-generation at proof lines 2607-2609. Give the step from the fourth description to the third.

Source reference. Original lemma 05GH: sources/stacks-chow/tags.txt, source lines 7061–7061. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0AIK: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 2453–2453. Pointer outside the counted span, invoked once

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for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0AIC: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 2135–2135. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0GXF: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 1633–1633. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0GXH: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 1714–1714. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original remark 0AIS: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 2769–2769. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Definition 0.1. Let S be a scheme. We say a sheaf X on $(Sch/S)_{fppf}$ is an *affine formal algebraic space* if there exist

(1) a directed set Λ,
(2) a system $(X_\lambda, f_{\lambda\mu})$ over Λ in $(Sch/S)_{fppf}$ where
   (a) each X_λ is affine,
   (b) each $f_{\lambda\mu} : X_\lambda \to X_\mu$ is a thickening,

such that

$$X \cong \operatorname{colim}_{\lambda\in\Lambda} X_\lambda$$

as fppf sheaves and X satisfies a set theoretic condition (see Remark 0AIS). A *morphism of affine formal algebraic spaces* over S is a map of sheaves.

Lemma 0.2. *Let S be a scheme. Let X be an affine formal algebraic space. The following are equivalent*

(1) *X is Noetherian,*
(2) *X is adic* and for every closed immersion $T \to X$ with T a scheme, T is Noetherian,*
(3) *X is adic* and for some choice of $X = \operatorname{colim} X_\lambda$ as in Definition 0.1 the schemes X_λ are Noetherian, and*
(4) *X is weakly adic and for some choice $X = \operatorname{colim} X_\lambda$ as in Definition 0.1 the schemes X_λ are Noetherian.*

Proof. Assume X is Noetherian. Then $X = \operatorname{Spf}(A)$ where A is a Noetherian adic ring. Let $T \to X$ be a closed immersion where T is a scheme. By Lemma 0AIC we see that T is affine and that $T \to \operatorname{Spec}(A)$ is a closed immersion. Since A is Noetherian, we see that T is Noetherian. In this way we see that (1) $\Rightarrow$ (2).

The implications (2) $\Rightarrow$ (3) and (2) $\Rightarrow$ (4) are immediate (see Lemma 0AIK).

To prove (3) $\Rightarrow$ (1) write $X = \operatorname{Spf}(A)$ for some adic ring A with finitely generated ideal of definition I. We are also given that the rings A/I_λ are Noetherian for some

fundamental system of open ideals I_λ. Since I is open, we can find a λ such that $I_\lambda \subset I$. Then A/I is Noetherian and we conclude that A is Noetherian by Algebra, Lemma 05GH.

To prove (4) $\Rightarrow$ (3) write $X = \mathrm{Spf}(A)$ for some weakly adic ring A. Then A is admissible and has an ideal of definition I and the closure I_2 of I^2 is open, see Lemma 0GXF. We are also given that the rings A/I_λ are Noetherian for some fundamental system of open ideals I_λ. Choose a λ such that $I_\lambda \subset I_2$. Then A/I_2 is Noetherian as a quotient of A/I_λ. Hence I/I_2 is a finite A-module. Hence A is an adic ring with a finitely generated ideal of definition by Lemma 0GXH. Thus X is adic* and (3) holds. $\square$